

EXPERIMENT 7

RANDOM FORESTS – OOB ERROR, BAGGING AND FEATURE SUBSAMPLING

Dataset:

John_Wick_Universe_Synthetic_Dataset.csv

Dataset Shape:

100000 rows × 12 columns

Target:

status

Training Rows:

80000

Testing Rows:

20000

Encoded Features:

62

Missing Values:

0

Duplicate Rows:

4

DECISION TREE

Accuracy: 0.3376

BAGGING

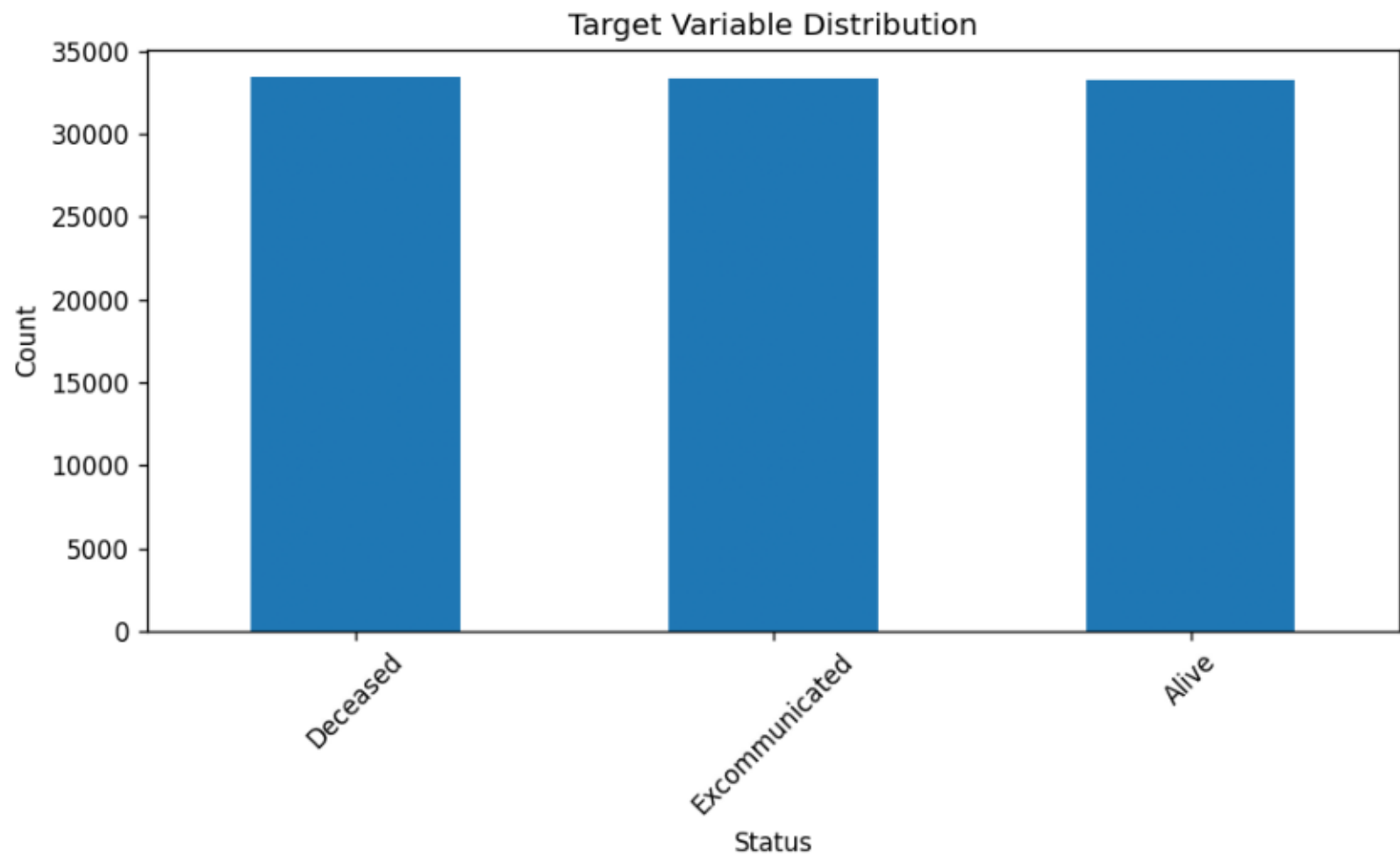
Trees: 20

Accuracy: 0.3324

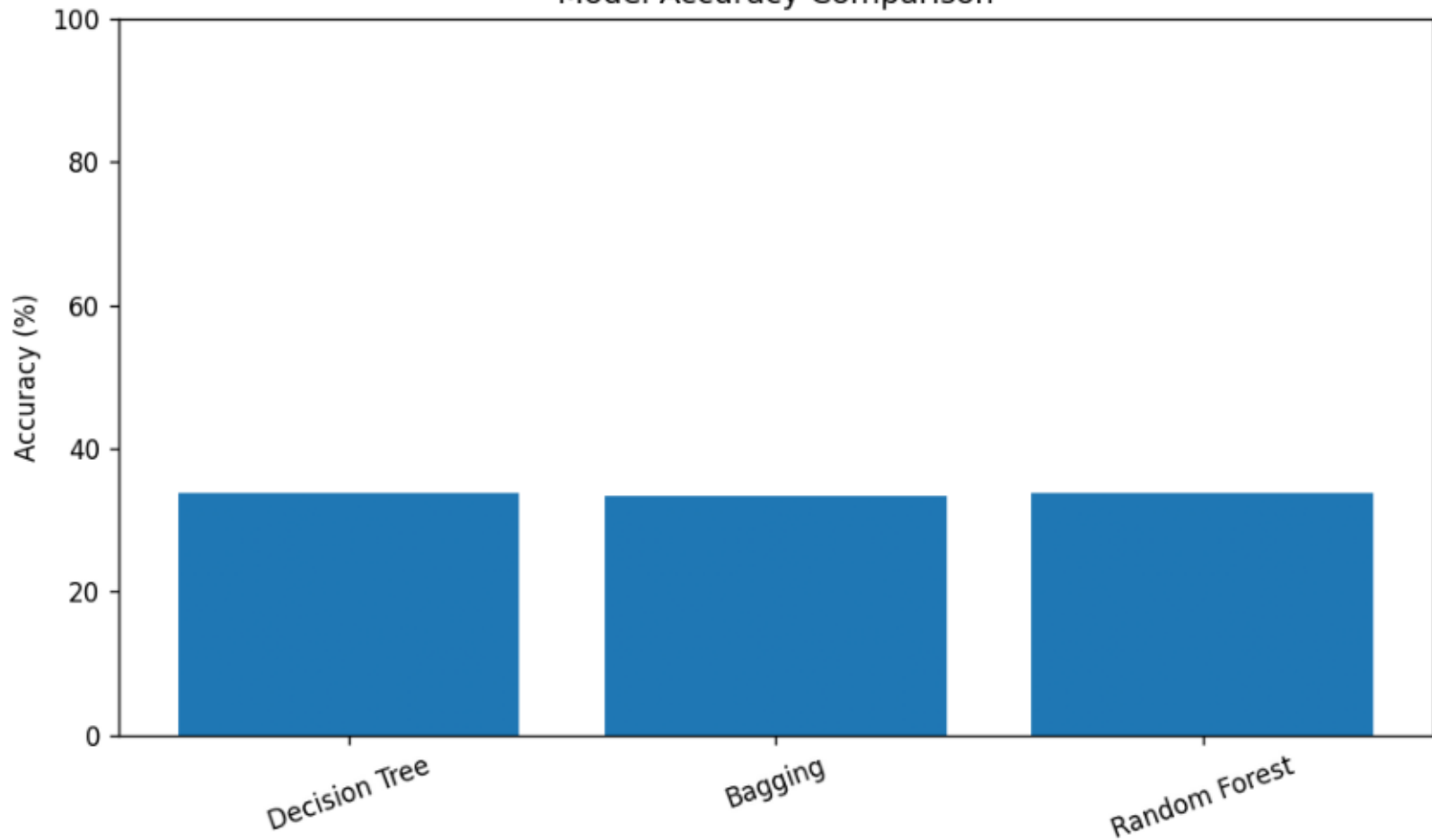
OOB Accuracy: 0.3345

OOB Error: 0.6655

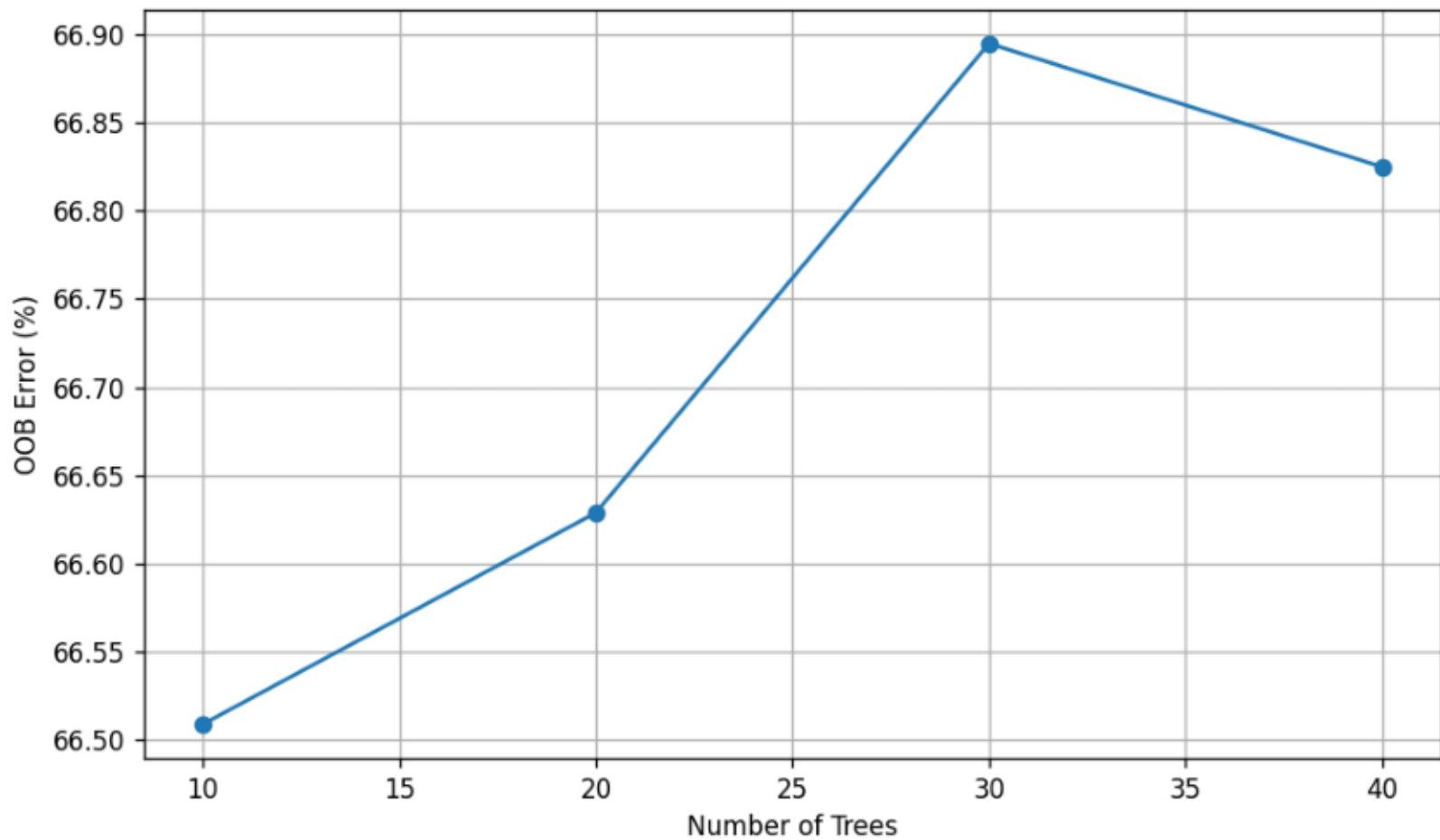
RANDOM FOREST



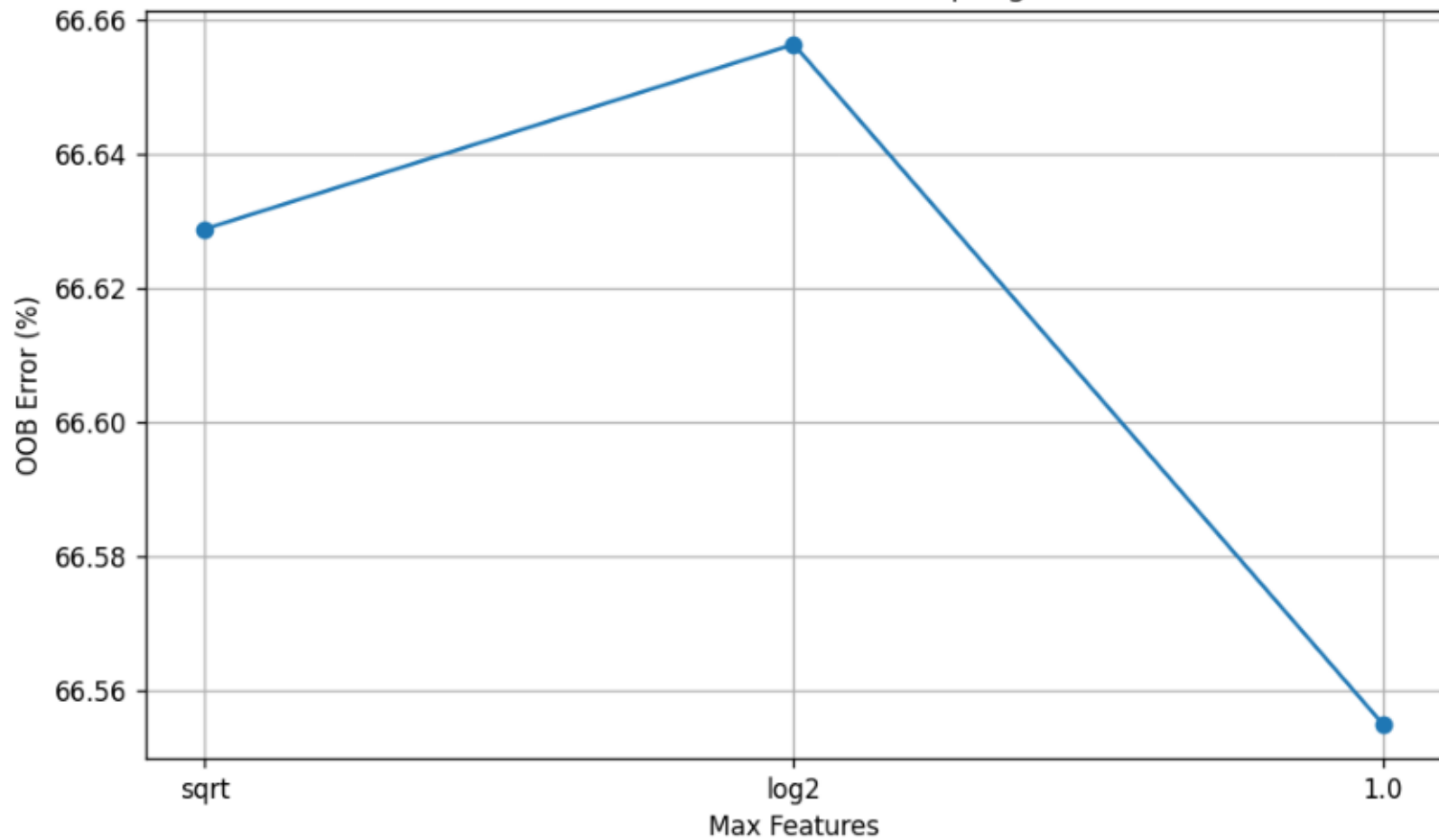
Model Accuracy Comparison



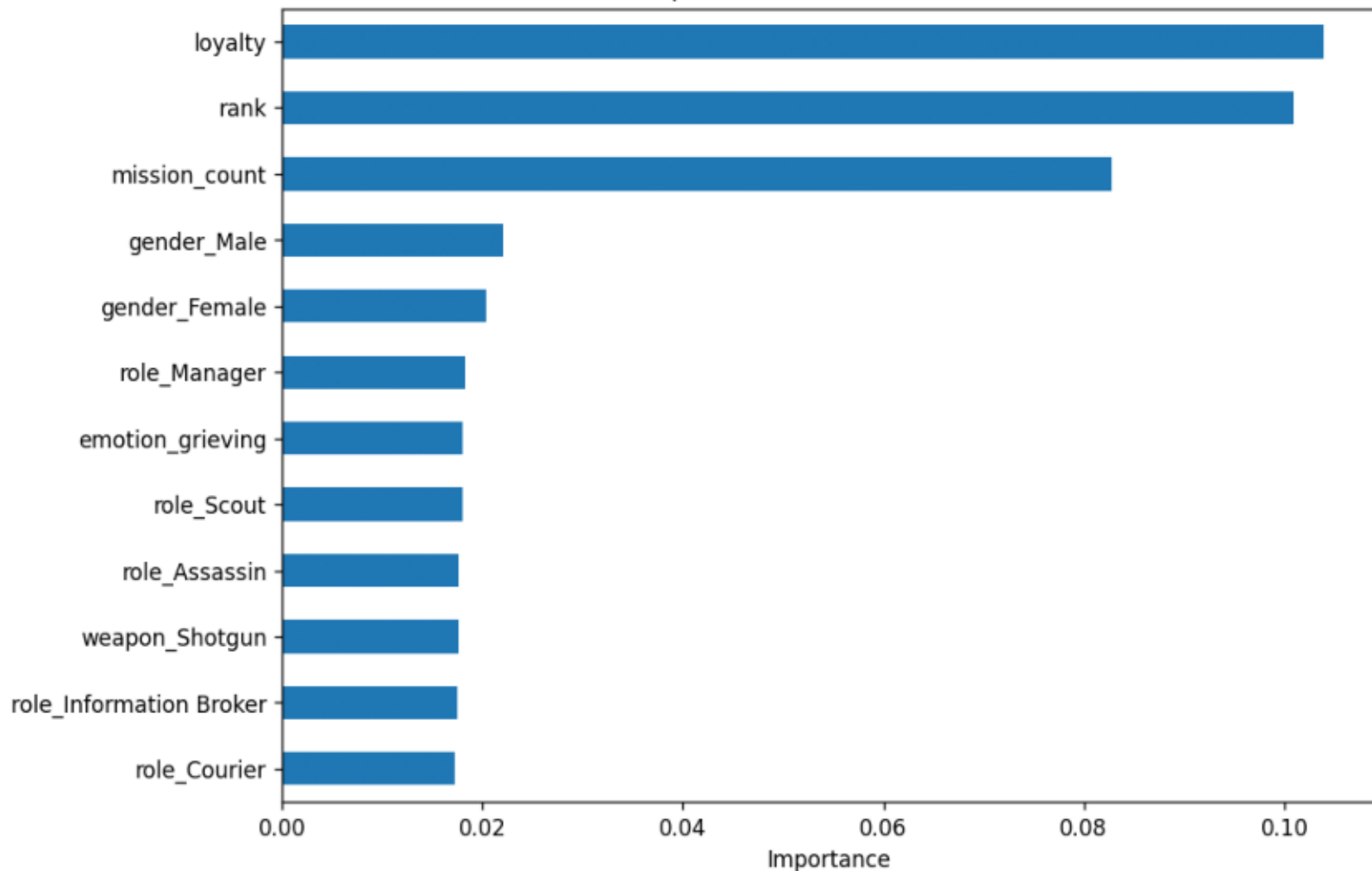
OOB Error vs Number of Trees



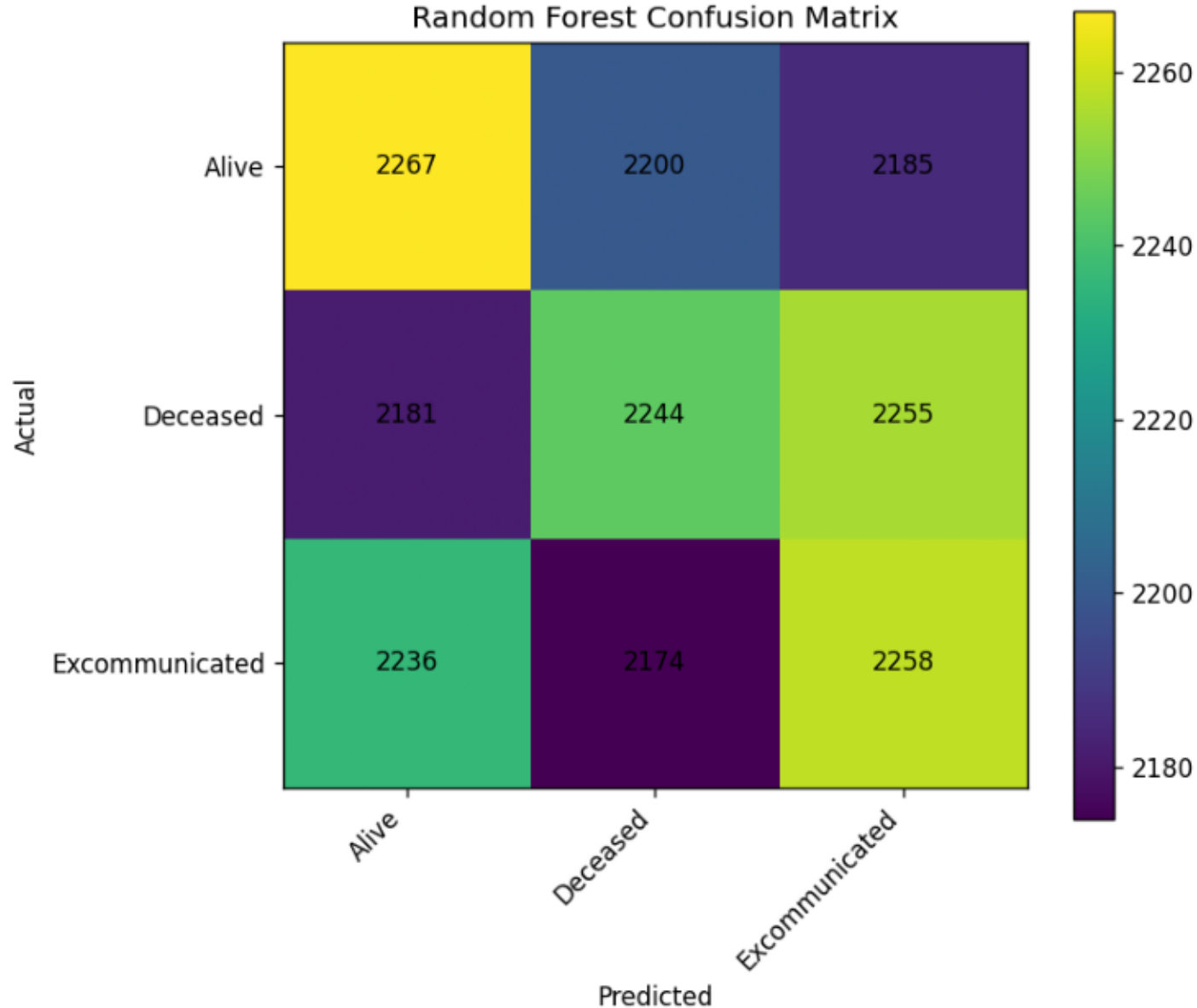
Effect of Feature Subsampling



Top Random Forest Features



Random Forest Confusion Matrix



Numeric Feature Correlation

